How our grid model works, from start to finish

A plain-language walk-through for the call with Juan. Every number comes from runs logged in docs/provenance.md (core_grid_dcopf, 26–27 October 2026).

The one thing to remember

We use **two different pictures of where a country's power enters the grid**, in two different places:

Where	How a country's exchange enters	Why
Computing the transfer limits (N-0 and N-1, step 3)	**Spread** evenly over the country's 380 kV buses	Closer to reality and to what grid operators do; one bus makes the lines next to it look full too early
Everything that runs on the reduced network: branch limits (step 4), Tulipa (5a), the 2025 test (5b)	**One bus** per country (the reduced network has one node per country)	That is what a reduced network is: one point per country

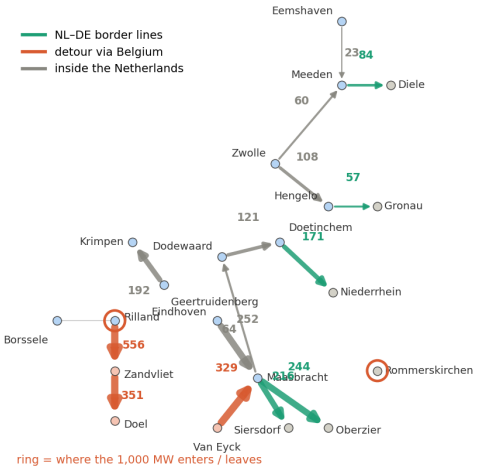
So the spread is only used where the transfer limits are calculated. That one calculation covers both the intact grid (N-0) and the outages (N-1). The reduced network that Tulipa runs on still routes every country from one bus. That mismatch is a known limitation (section 7).

The same example drawn: where the 1,000 MW flows

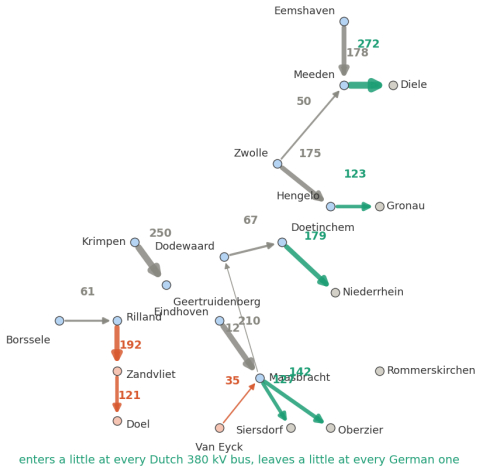
Left: all 1,000 MW enters at Rilland and leaves at Rommerskirchen (the September approach). Right: it enters a little at every Dutch 380 kV bus and leaves a little at every German one (our approach). Arrow width and number = MW on that line. Green = Dutch–German border lines, orange = the detour through Belgium, grey = lines inside the Netherlands. Station positions are schematic.

Where 1,000 MW from the Netherlands to Germany flows (today's grid, no outage, schematic positions)

One bus: all 1,000 MW from Rilland to Rommerskirchen
the September approach (numbers: MW on the line)

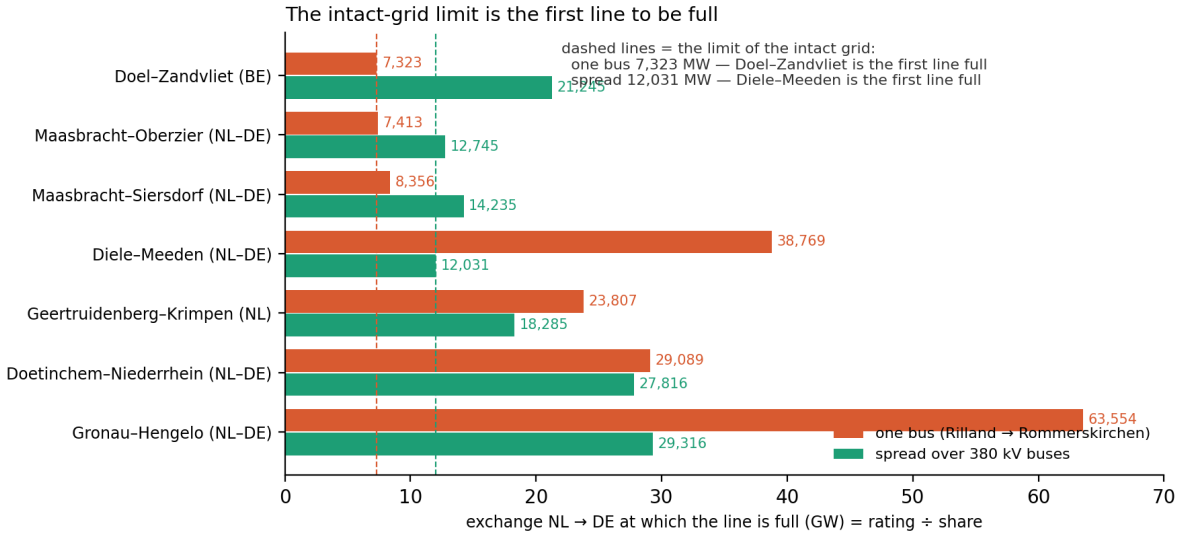


Spread: 1,000 MW from all 33 Dutch to all 324 German 380 kV buses
our approach since 13 October (numbers: MW on the line)



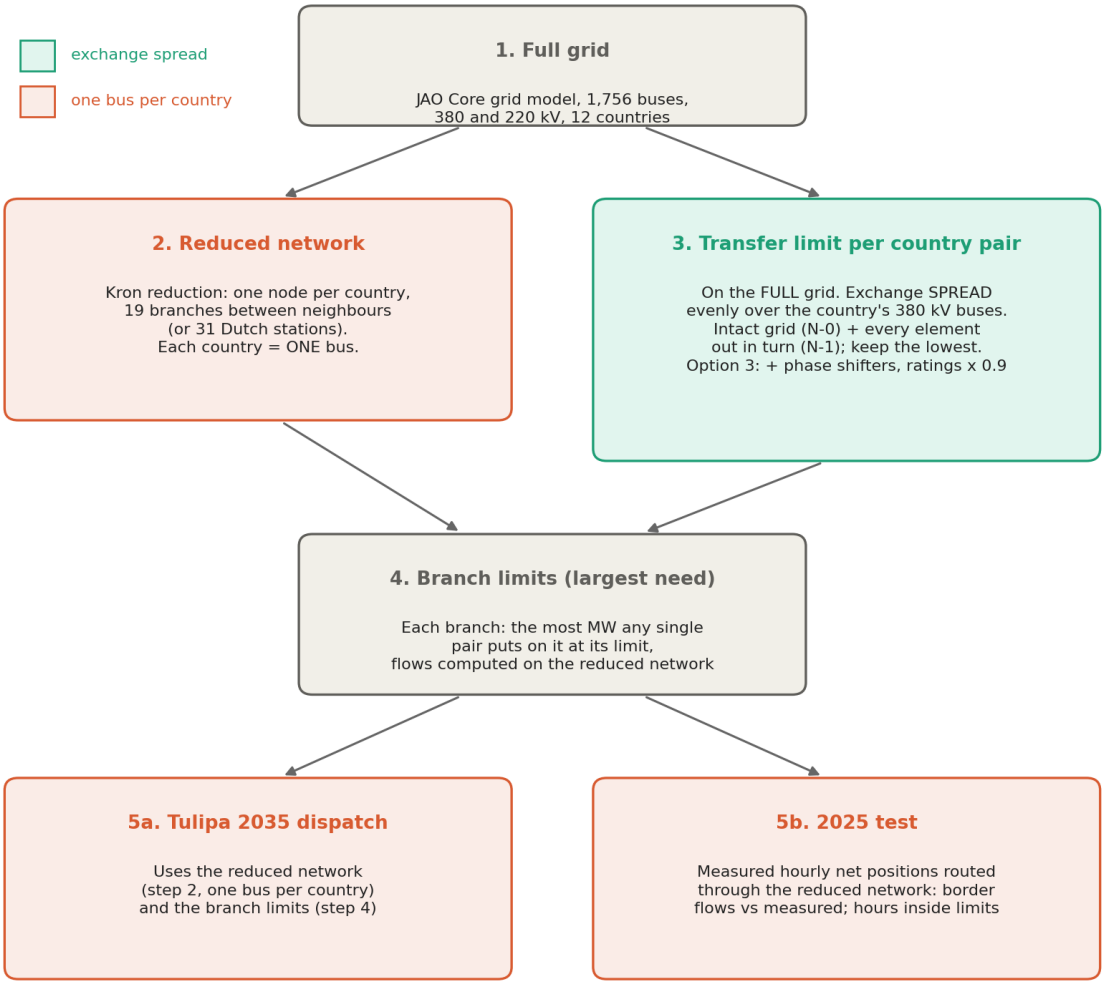
Direct over the NL–DE border: one bus 773 MW, spread 843 MW; via Belgium: 227 vs 157 MW. Totals cross-checked: the five NL–DE lines sum to 772 / 843 MW.

With one bus, 556 MW leaves Rilland towards Zandvliet and 351 MW continues on Doel–Zandvliet inside Belgium: the Belgian line carries 35 % of the exchange and is the first to be full. With the spread, the north carries its share (Eemshaven–Meeden 178 MW, Meeden–Diele 272 MW) and the Belgian detour shrinks to 121 MW on Doel–Zandvliet.



Each bar is one line's limit for this exchange: its rating divided by its share. The transfer limit of the intact grid is the shortest bar: 7,323 MW with one bus (Doel–Zandvliet), 12,031 MW with the spread (Diele–Meeden). The N-1 limit repeats this chart once per outage and keeps the shortest bar of all.

The process in one picture



1. The full grid

JAO's Core Static Grid Model (March 2026 release): every 380 kV and 220 kV line and transformer of the 12 Core countries, 1,756 buses. We added missing Dutch lines (from GridAware and TenneT data) and two assumed Vierverlaten transformers. Each line's rating is √3 × voltage × current, with the current the average of JAO's seasonal values. All calculations are DC load flows (no losses, no voltages).

2. The reduced network

Tulipa cannot run on 1,756 buses, so the grid is shrunk (Kron reduction) to one node per country, kept at one chosen bus (the representative bus; for the Netherlands Rilland, for Germany Rommerskirchen). Only branches between neighbouring countries are kept: 19 branches. Each branch gets a reactance, so power still splits over parallel paths roughly as on the full grid. A second version keeps 31 Dutch stations as separate nodes. What this network does NOT have yet is a limit per branch; that comes from steps 3 and 4.

Where we came from: the method over time

When	Method	Exchange enters at	Lines that may set the limit
September	**The original approach.** Intact grid only (N-0): the reduction package's transfer limits between the representative buses; largest need → branch limits	one bus per country	any line or transformer
early October	Juan's N-1 copied from the MATLAB file: border lines switched off in turn	one bus	any line or transformer
8–12 October	Investigation: why are the N-1 limits so tight? (archived scripts 78–82)		
13 October	**Two fixes at once**: only 380 kV lines may set a limit, and the exchange is spread over the country's 380 kV buses	**spread**	380 kV lines
14 October	Option 3: every element switched off in turn, phase shifters, 10 % margin	spread	380 kV lines
16 October	1 km rule	spread	380 kV lines ≥ 1 km
26 October	Decomposition for the note to Juan: everything redone one change at a time		

So the spread came in on 13 October, together with the rule on which lines may set a limit. From then on every limit calculation uses it, intact grid included. The original September approach is not used anywhere anymore; its numbers survive only as the first row of the decomposition table (section 6), as the starting point against which the changes are measured. Its one idea — a line's limit is rating ÷ share, take the smallest over all lines — is still the heart of every entry in today's calculation.

Which grid each method uses for the transfer limit

The transfer limit is always calculated on the **full grid**: in the reduction package, in Juan's MATLAB file and in ours. The small 12-node network is never used to calculate a transfer limit; it only comes in afterwards (largest need, Tulipa, 2025 test). What differs is which of the full grid's buses are used as start and end points:

	Reduction package	Juan's MATLAB	Ours
Grid the flows are computed on	full grid, 1,756 buses, every line	full grid	full grid
Where country A injects	**1 bus** (its representative)	**1 bus**	**all of A's 380 kV buses** (33 for NL)

	Reduction package	Juan's MATLAB	Ours
Where country B withdraws	1 bus	1 bus	all of B's 380 kV buses (324 for DE)
Outages	none	border lines	every line and transformer
Lines that may set the limit	every line	every line	380 kV lines of at least 1 km (all lines carry flow)

The package and the MATLAB file use only 12 of the 1,756 buses as start and end points: one per country, the same buses that later become the nodes of the small network. That is why they can feel as if they work in the small network. They do not: the flows run through the whole grid; only the start and end points are 12 buses. (The package even computes a limit for every pair of buses, about 1.5 million, and then keeps only the 66 between representative buses.)

How N-0 and N-1 work together: one list, keep the smallest

It is one procedure, run once per pair of countries (66 pairs), and it produces a list of numbers:

Entry	Grid state	Netherlands → Germany, today's grid, spread
0	intact grid (this is what people call N-0)	12,031 MW, set by Diele–Meeden
1	one Meeden–Diele circuit switched off	6,629 MW, set by the other Diele–Meeden circuit
2	Doel–Zandvliet circuit switched off	12,022 MW
3	Maasbracht–Oberzier switched off	8,974 MW, set by Maasbracht–Siersdorf
4	Geertruidenberg–Krimpen circuit switched off	10,070 MW
…	… every one of the 2,962 elements in turn	…
limit	**the smallest entry in the list**	**6,629 MW** (entry 1)

Each entry is the same calculation: spread the exchange, work out every monitored line's share, each line is full at rating ÷ share, take the smallest line. Entry 0 does it on the intact grid; every later entry does it with one element removed (an exact shortcut updates the shares instead of solving the grid again). Nothing is merged or averaged: the N-1 limit is simply the smallest number in the list. If no outage were worse than the intact grid, the N-1 limit would equal the N-0 one. For the Dutch–German pair the worst case is losing one of the two Meeden–Diele circuits: the flow moves onto the remaining circuit, which is full at 6,629 MW instead of 12,031.

How the spread and the PTDFs fit together

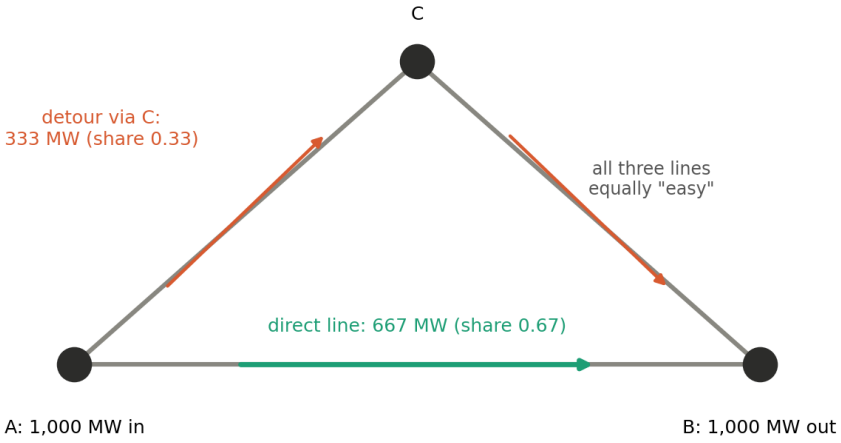
A PTDF (share) is never a property of a line alone; it is the share of a line **for a given start and end of the exchange**. Change the start or end and every share changes. So "the PTDFs" always means "the PTDFs of exchange X", and the spread only says what X is.

	Exchange X	Diele–Meeden share	How it is computed
One bus	1 MW from Rilland to Rommerskirchen	0.084	one grid solve
Spread	1/33 MW from each Dutch 380 kV bus to 1/324 MW at each German one	0.272	the same solve; because the grid is linear, the spread share is the average of the 33 start shares minus the average of the 324 end shares

This is exactly how JAO computes the zone-to-zone PTDFs it publishes: bus shares weighted with the operator's shift key. Our key is equal weights. Two different sets of shares live in our chain: the shares of the 2,396 full-grid lines (spread; used for the limits) and the shares of the 19 reduced branches (one node per country; used for the largest-need step, Tulipa and the 2025 test). The reduced network cannot use the spread because there a country is a single node: there is nothing to spread over.

How a share (PTDF) is calculated

Every step above needs the same basic number: if 1 MW goes from A to B, how much of it flows on each line? That number is the line's **share** (PTDF). Power cannot be steered: it takes every possible path, and more of it takes the easier paths. "Easy" is the line's reactance (low reactance = easy).



Three buses, three identical lines. From A to B there are two paths: the direct line (one line) and the detour via C (two lines in a row, so twice as hard). Power splits in inverse proportion: two thirds direct, one third via C. So the direct line's share is 0.67, the detour lines' 0.33.

The computer does exactly this for the whole grid at once. Recipe: (1) put +1 MW where the exchange starts and −1 MW where it ends; (2) solve one set of linear equations for the whole grid (the DC load flow: each line's flow is the difference in "voltage angle" between its two ends divided by its reactance, and at every bus what flows in must flow out); (3) read the flow on every line; that is its share. One solve gives the shares of all lines for that exchange.

Where the exchange starts matters. One bus: +1 MW at Rilland, −1 MW at Rommerskirchen. Spread: +1/33 MW at each of the 33 Dutch 380 kV buses and −1/324 MW at each of the 324 German ones. Because everything is linear, the spread share is simply the average of the shares you would get from each bus on its own. **N-1:** switch one line off and solve again; we use an exact shortcut formula that updates the solution instead of solving from scratch (thousands of outages in seconds).

Two kinds of shares in our model. In step 3 the shares come from the **full grid** (1,756 buses, spread). In steps 4 and 5 they come from the **reduced network**: the same recipe on only 12 nodes and 19 branches, where each country is one node. The reduced network's reactances come from the Kron reduction, which is built so that it splits power like the full grid when each country injects at its **one** representative bus.

3. Transfer limits between countries (where N-0, N-1 and the spread live)

For each pair of countries (66 pairs) we ask, on the full grid: how many MW can go from A to B before some line is full? A line is full when exchange × its share = its rating, so each line gives a limit = rating / share, and the transfer limit is the smallest of these over the lines that may set a limit. The share (PTDF) depends on where the exchange starts and ends; this is where the spread comes in.

Example: 1,000 MW from the Netherlands to Germany, intact grid.

	One bus (Rilland to Rommerskirchen)	Spread (33 Dutch, 324 German 380 kV buses)
Goes straight over the NL–DE border	773 MW	843 MW

	One bus (Rilland to Rommerskirchen)	Spread (33 Dutch, 324 German 380 kV buses)
Detours through Belgium	227 MW	157 MW
First line to be full	Doel–Zandvliet in Belgium, at about 7,300 MW	Diele–Meeden (NL–DE tie), at about 12,000 MW
Transfer limit, intact grid / with N-1	7,323 / 4,544 MW	12,031 / 6,629 MW

With the whole export starting at Rilland, right next to Belgium, a quarter of it detours through Belgium and a Belgian line fills first. Spread over the whole country, more of it crosses directly and the real Dutch–German tie fills first. The spread does not push more power through the same lines; it changes which line takes the biggest share of each MW.

N-1 means: switch off one element, recompute, see which line is full first; repeat for every element; keep the lowest limit. There are two separate lists, and they are easy to mix up:

List	Meaning	Juan's MATLAB file	Ours
Outages	what is switched off, one at a time	only lines crossing a border	every line and transformer (2,962)
Monitored lines	what is allowed to set the limit	every line and transformer	only 380 kV lines of at least 1 km

Option 3 (the limits we use) adds two things in this same step: the phase shifters on 8 corridors are set for each exchange within their range (one setting that must hold under every outage), and every rating is reduced by 10 % (the reliability margin JAO also uses).

4. From pair limits to branch limits: the largest need

We now know safe limits per pair of countries, but Tulipa needs one number per line of the small network. For each line we ask, pair by pair: if this pair runs at its safe limit, how many MW land on this line? The line gets the largest of those answers. Real numbers for the Netherlands–Germany line (today's grid, our N-1):

Pair at its safe limit	Safe limit	Share on the NL–DE line	MW it needs on the NL–DE line
Austria → Netherlands	6,825 MW	0.632	**4,316 MW (the largest)**
Germany → Netherlands	6,629 MW	0.632	4,193 MW
Czechia → Netherlands	5,194 MW	0.632	3,285 MW
France → Netherlands	6,885 MW	0.459	3,162 MW
… 62 more pairs			less

So the NL–DE line gets 4,316 MW. Why the largest and not the smallest: with less room, Tulipa could never let Austria → Netherlands reach its safe 6,825 MW. With the largest, every pair on its own can reach its safe limit. The price: Germany → Netherlands can now reach 4,316 / 0.632 = 6,829 MW in Tulipa, a bit above its own safe 6,629, and when several pairs trade at once their needs add up on the same line without any check (for 2035 the Belgian ties allowed about 20 % too much, fixed with a tie cut). The shares here come from the small network, so every country is one bus again. In the code and in the reduction package this step is called the "envelope".

5. How we check it: the 2025 test

We take every hour of 2025. Each country's net position is computed from the measured hourly physical cross-border flows (ENTSO-E), with the ALEGrO link taken from JAO. We push those positions through the reduced network and compare, hour by hour:

Check	Result (country network, option 3)
Model border flows vs measured physical flows, 19 Core borders	mean hourly correlation 0.90; average error 0.39 GW per border
Belgium–Netherlands	bias +0.33 GW (model too much towards NL), average error 0.59 GW
Hours in which every branch stays within its limit ("hours inside")	91.7 %
Our pair limits vs what JAO offered the market in 2025 (MaxBEX)	median 1.09 times JAO
Measured 2025 flows inside our limits	45.5 % of hours (84.3 % without Croatia–Hungary and Poland–Slovakia)

So the answer to Juan's question ("can the flows in the reduced network be compared with measured physical cross-border flows, hourly?") is yes: that is exactly this test, at hourly resolution.

6. What changed against Juan's method, one step at a time

Same network, same 2025 test; each row adds one change.

Step	Hours inside	NL–DE limit	BE–NL limit
Intact grid only, one bus	83.5 %	7,323 MW	5,799 MW
Juan's N-1 as written	36.1 %	4,968 MW	5,517 MW
+ every element can fail, failed circuit's rating removed	14.4 %	4,544 MW	3,599 MW
+ only 380 kV lines of at least 1 km may set the limit	52.6 %	4,544 MW	3,599 MW
+ exchange spread over 380 kV buses = our N-1	95.3 %	6,629 MW	3,724 MW
+ phase shifters and 10 % margin = option 3 (used)	91.7 %	8,561 MW	5,017 MW

Spreading is the change that matters most. It is not about a badly chosen bus: with the "best" single bus per country the result is no better (52.9 % with the Netherlands at Diemen), because whichever bus takes the whole exchange, the lines next to it fill first.

Why we think the spread is right: grid operators do the same thing with their generation shift keys (the current Core rule: shares in proportion to generation and load per node, ACER Decision 10/2026). And it matches JAO's published line sensitivities best: over 80 hours of 2026, 45 % of our line shares are within 25 % of JAO's (69 % on TenneT NL lines), against 26 % (32 %) with one bus. Spreading over all buses including 220 kV does worse (37 %), as does a Dutch generation + load key (42 %).

7. Limitations to keep in mind

Limitation	What it means
Routing and limits use different pictures (spread vs one bus)	Loop flows in Tulipa are off; BE–NL by +0.3 to +0.7 GW on average. A fix was tested, not adopted (next page).
Limits checked one exchange at a time	Several exchanges together can exceed the secure value; patched with a tie cut on the Belgian ties.
Equal spread, not the operators' real keys	Good on average, weak further east (below 30 % of lines within 25 % for AT, CZ, HU, HR, SI, RO).

Limitation	What it means
Only 380 kV may set a limit	Fits the Netherlands; misses 220 kV limits between Bavaria and Austria (30 % of 2025 Core shadow prices were on 220 kV).
DC load flow, one rating per line, tuned on 2025 only	No losses or voltages; no summer/winter ratings; no test on another year.

The tested fix: make the reduced network behave like the spread

The problem in one sentence: the limits are calculated with the exchange spread, but the reduced network (Tulipa, 2025 test) routes each country from one bus. The fix we tested changes the reduced network so that it routes like the spread too.

Picture it as knobs. Each of the reduced network's branches has one knob: its reactance (how "easy" it is). Kron reduction sets the knobs so the small network copies the full grid when each country injects at its one bus. The fix turns the knobs again, so the small network copies the full grid when each country injects spread out.

How the knobs are turned. (1) On the full grid, inject 1 MW spread over one country (or at one Dutch station) and record how much crosses each border, for every node. (2) Do the same on the reduced network. (3) Let the computer turn the knobs until the reduced numbers are as close as possible to the full-grid numbers (least squares). (4) Recompute the branch limits (step 4) with the new shares. Script: scripts/42_spread_ptdf_fit_nodal.py (42b weights each node by how much power it injects in 2025).

Tested on today's grid, Germany split in two, 31 Dutch stations	Before (Kron)	Fitted	Fitted, weighted	Full grid, all spread (best possible)
Error of the shares, per branch (RMS, MW per MW)	0.053	0.014	0.020	0
2025 routing error, 19 borders (GW)	0.435	0.318	0.304	0.275
Belgium–Netherlands bias (GW)	+0.54	+0.52	+0.41	+0.16
NL→BE: reduced network allows / full grid secure	×1.20	—	×1.44	—

Why we did not adopt it. It clearly improves the routing, especially on the eastern borders, but (a) Belgium–Netherlands improves little, because one node per German half cannot copy the whole loop through the Netherlands; (b) with the new shares the Belgian ties allow even more than is secure (×1.44 instead of ×1.20), so the tie cut would have to be redone and checked; (c) the weighted version is tuned on 2025 data. We chose to freeze the model and list this as a limitation.

8. Cheat sheet for the call with Juan

Juan says / asks	Answer
"You spread the exchange to all nodes equally when calculating the PTDFs and the TTCs?"	Only for the transfer limits (TTCs), and only over each country's 380 kV buses, equally; not all nodes. The reduced network's own PTDFs (what Tulipa and the 2025 test use) still have one bus per country. We tested a fit that makes them follow the spread (routing error 0.435 → 0.30 GW) but did not adopt it.
"The ~95 % in 2025"	That is our N-1 alone. The limits we actually use (with phase shifters and the 10 % margin) give 91.7 %.
"Interesting even for N-0"	Yes, it is the same calculation: intact-grid NL–DE goes from 7,323 to 12,031 MW, and the limiting line moves from Belgium to the NL–DE tie.
"Can I share it with German colleagues?"	Fine to share our method, scripts and results (your call). It is the simplest form of the shift keys every TSO uses, which makes it easy to defend. Raw JAO files cannot be passed on.
"Plans for 150 and 110 kV?"	Yes: a full Dutch load flow after the Tulipa dispatch, on GridAware's network, to show congestion per station. We miss demand and generation per 110/150 kV station and the normally-open points. Does GridAware have these?
"IP26 additions"	Already in our 2035 380/220 kV grid (Amir's package, two selection rules). Not yet in the 150/110 kV grid.

Juan says / asks	Answer
"Compare reduced-network flows with measured physical flows, hourly?"	Yes, that is our 2025 test: hourly, 19 Core borders, correlation 0.90, average error 0.39 GW per border.

9. Words

Word	Meaning
PTDF / share	MW on a line per MW exchanged between two countries
Transfer limit (TTC)	Largest exchange A→B with no monitored line overloaded, intact grid and every single outage
N-0 / N-1	Intact grid / every element switched off in turn, keep the worst
Shift key (GSK)	Where an exchange enters and leaves the grid inside a country
Kron reduction	Shrinking the grid to a few nodes while keeping how power splits between them
Largest need ("envelope" in the code)	Branch limit = the most MW any single pair of countries puts on the branch when it runs at its transfer limit
Option 3	Our final limits: N-1 + phase shifters + 10 % margin
Hours inside	Share of 2025 hours in which every branch stays within its limit
MaxBEX	The largest exchanges JAO offered the market (flow-based)